

Residual Analysis

Objective

Residual analysis was conducted to evaluate the performance of the selected multiple regression model. The objective is to determine how accurately the model predicts monthly sales and identify stores where the prediction error is unusually large.

Residual analysis also helps detect patterns that may indicate missing variables, unusual store performance, or limitations of the regression model.

Final Regression Model

The selected model predicts **Monthly Sales** using multiple business factors, including numerical variables and dummy variables for categorical data.

General equation:

$$\text{Predicted Monthly Sales} = \beta_0 + \beta_1(X_1) + \beta_2(X_2) + \beta_3(X_3) + \beta_4(\text{Dummy Variable})$$

where:

- X_1 = Marketing Spend
 - X_2 = Footfall
 - X_3 = Inventory Availability Percentage
 - Dummy Variable = Region or Store Type
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Predicted Sales

Predicted Monthly Sales were calculated for every record using the final regression equation.

For each observation:

$$\text{Predicted Sales} = \text{Regression Equation Output}$$

These predicted values represent the model's estimate of expected monthly sales.

Residual Calculation

Residuals were calculated using the following formula:

$$\text{Residual} = \text{Actual Monthly Sales} - \text{Predicted Monthly Sales}$$

Interpretation:

- **Positive Residual:** Actual sales were higher than predicted (the model under-predicted sales).
- **Negative Residual:** Actual sales were lower than predicted (the model over-predicted sales).
- **Residual close to zero:** The model predicted sales accurately.

Records with the Largest Positive Residuals

The following records showed the largest positive residuals:

Record ID	Actual Sales	Predicted Sales	Residual	Business Interpretation
---	---	---	---	Sales exceeded expectations, suggesting additional positive factors not included in the model.
---	---	---	---	Store performed better than predicted.
---	---	---	---	Possible impact of local promotions or seasonal demand.
---	---	---	---	Higher customer demand than expected.
---	---	---	---	Potential operational strengths not captured by the regression variables.

Records with the Largest Negative Residuals

The following records showed the largest negative residuals:

Record ID	Actual Sales	Predicted Sales	Residual	Business Interpretation
---	---	---	---	Sales were lower than predicted, suggesting operational challenges.
---	---	---	---	Possible inventory shortages or reduced customer demand.
---	---	---	---	Local market conditions may have affected performance.
---	---	---	---	Temporary business disruptions may not be captured by the model.
---	---	---	---	Customer behaviour differed from expected patterns.

Business Interpretation of Residuals

Positive Residuals

Large positive residuals indicate stores that performed significantly better than expected.

Possible explanations include:

- Successful local marketing campaigns
- Strong store management
- Seasonal demand
- Special promotions
- Customer loyalty
- Local events increasing footfall

These factors may not be represented in the regression model.

Negative Residuals

Large negative residuals indicate stores that performed worse than expected.

Possible explanations include:

- Stock shortages
- Operational inefficiencies
- Staff shortages
- Increased competition
- Local economic conditions
- Temporary disruptions

Again, these factors may not be included among the selected predictor variables.

Under-Prediction vs Over-Prediction

The residual analysis was reviewed to determine whether the model consistently under-predicts or over-predicts certain stores or customer segments.

Potential patterns include:

- Under-predicting high-performing stores
- Over-predicting low-performing stores
- Differences across regions
- Differences across store types

If a consistent pattern exists, additional explanatory variables should be considered in future models.

Model Performance Assessment

Overall, residuals should be randomly distributed around zero.

A random residual pattern suggests that the regression model is appropriate.

If residuals display systematic patterns, the model may violate regression assumptions or omit important business variables.

Limitations

Residual analysis indicates that:

- Regression cannot explain every variation in monthly sales.
 - External business factors such as local competition, economic conditions, weather, and promotional events may influence performance.
 - Some prediction errors are expected because not all relevant variables are available in the dataset.
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Conclusion

Residual analysis provides valuable insight into how well the regression model predicts monthly sales.

The model performs well when residuals are generally small and randomly distributed. Stores with large positive or negative residuals should be investigated individually to identify business factors not captured by the regression model.

Residual analysis supports continuous model improvement and helps leadership understand where predictions are reliable and where additional investigation may be required.